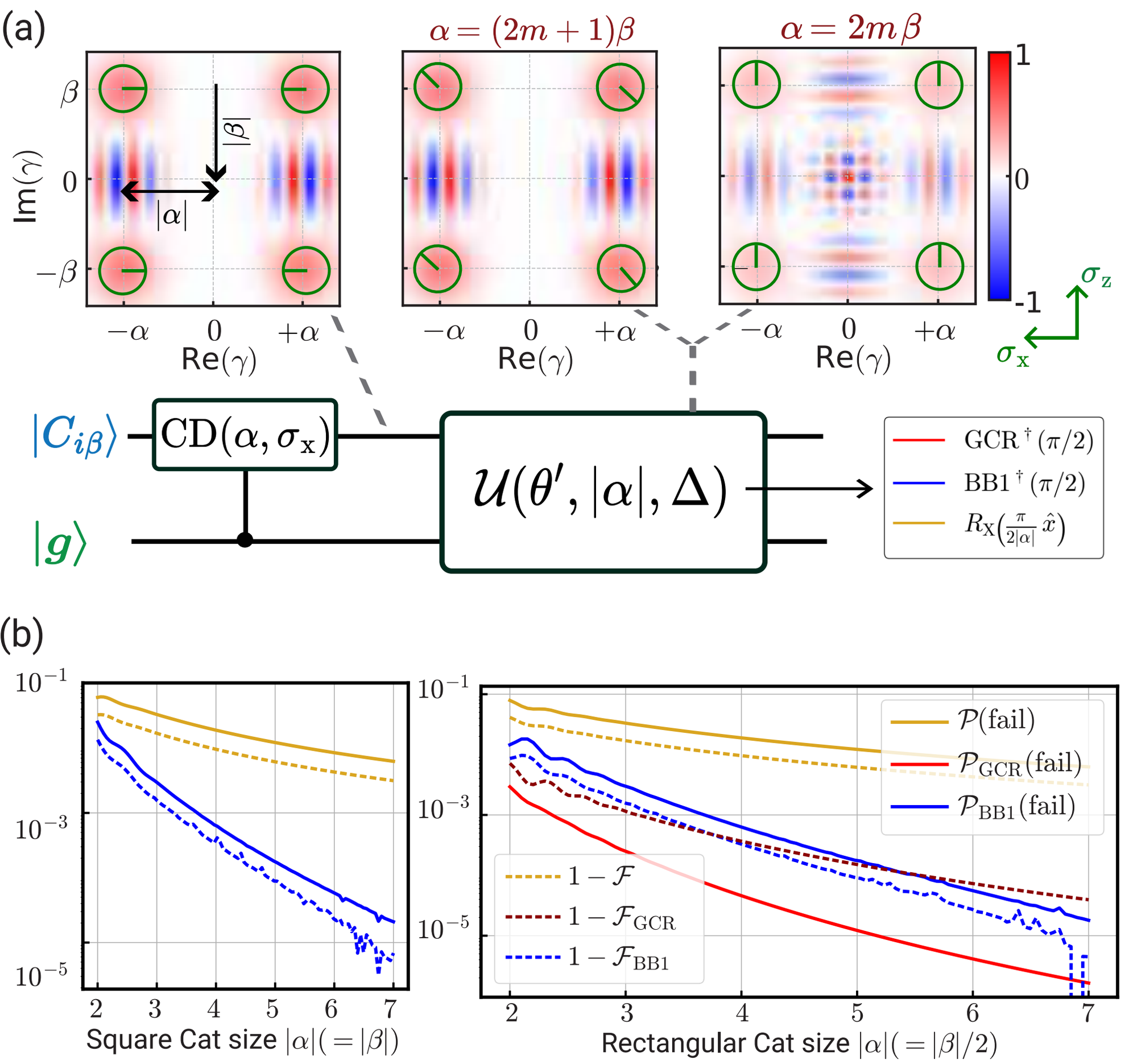

(a)
$\alpha = (2m+1)\beta$
$\alpha = 2m\beta$
β
0
$-\beta$
Im(γ)
$|\beta|$
$|\alpha|$
$-\alpha$
0
$+\alpha$
Re(γ)
1
0
-1
σ_z
σ_x
$|C_{i\beta}\rangle$
$|g\rangle$
CD(α, σ_x)
$\mathcal{U}(\theta', |\alpha|, \Delta)$
GCR$^\dagger(\pi/2)$
BB1$^\dagger(\pi/2)$
$R_X(\frac{\pi}{2|\alpha|}\hat{x})$
(b)
10^{-1}
10^{-3}
10^{-5}
2
3
4
5
6
7
Square Cat size $|\alpha|(=|\beta|)$
Rectangular Cat size $|\alpha|(=|\beta|/2)$
$\mathcal{P}$(fail)
$\mathcal{P}_{\rm GCR}$(fail)
$\mathcal{P}_{\rm BB1}$(fail)
$1-\mathcal{F}$
$1-\mathcal{F}_{\rm GCR}$
$1-\mathcal{F}_{\rm BB1}$